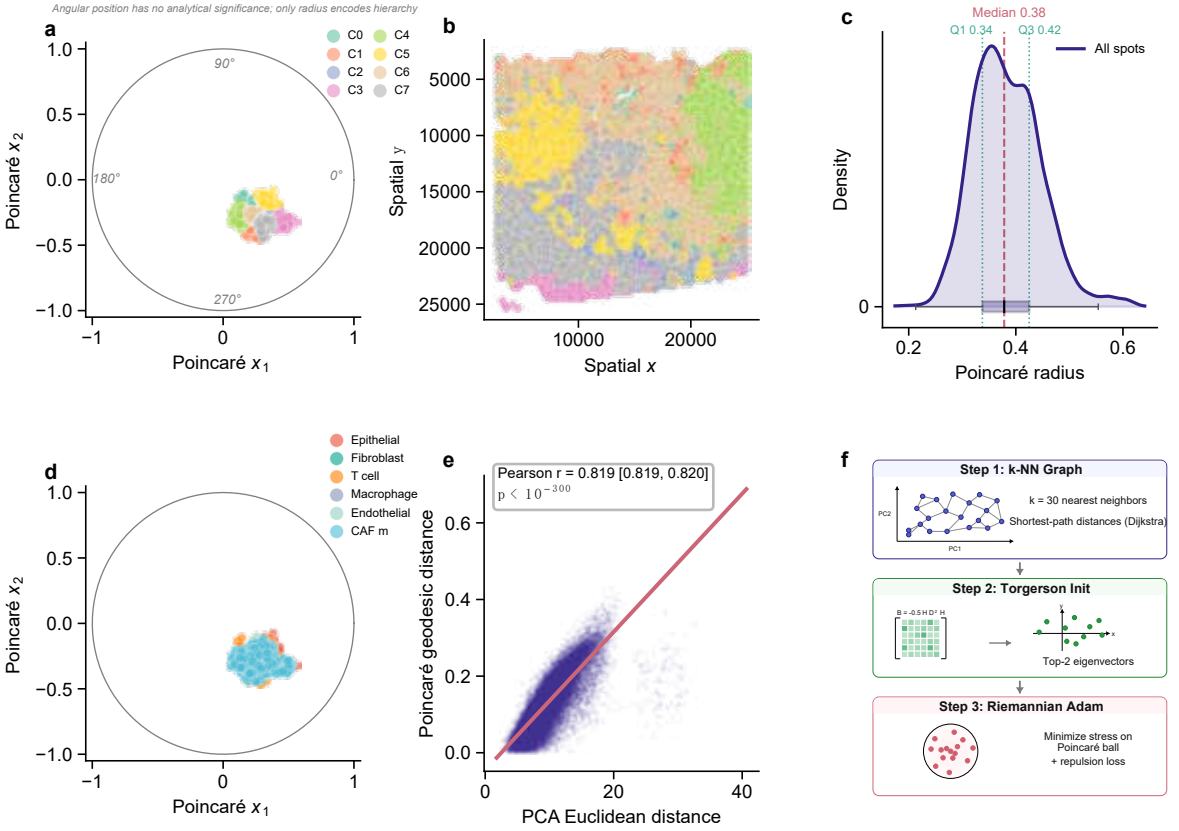
Angular position has no analytical significance; only radius encodes hierarchy
a
C0
C1
C2
C3
C4
C5
C6
C7
90°
180°
0°
270°
Poincaré x_2
Poincaré x_1
b
Spatial y
Spatial x
c
Median 0.38
Q1 0.34
Q3 0.42
All spots
Density
Poincaré radius
d
Epithelial
Fibroblast
T cell
Macrophage
Endothelial
CAF m
Poincaré x_2
Poincaré x_1
e
Pearson r = 0.819 [0.819, 0.820]
$p < 10^{-300}$
Poincaré geodesic distance
PCA Euclidean distance
f
Step 1: k-NN Graph
k = 30 nearest neighbors
Shortest-path distances (Dijkstra)
PC2
PC1
Step 2: Torgerson Init
$B = -0.5\, H\, D^2\, H$
Top-2 eigenvectors
Step 3: Riemannian Adam
Minimize stress on
Poincaré ball
+ repulsion loss